INTERNSHIP PROJECT REPORT

DESIGN AND IMPLEMENTATION OF A SMART MULTI-CELL BATTERY MANAGEMENT SYSTEM

Using ESP32, Wokwi Simulation and Blynk IoT

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A simulated and modular four-cell BMS prototype with supervisory protection, event-driven telemetry and cloud analytics.

Project Report Notes

Scope and evidence note

This report consolidates the uploaded Task 1–4 verification documents, the final Task 5–6 integration guide, the final firmware source and the dashboard / Serial Monitor screenshots supplied in November 2026. Test outcomes are described according to the evidence available. Temporary test fixtures and earlier verification builds are identified separately from the final integrated circuit and production firmware.

The prototype is a software-driven BMS demonstrator in Wokwi. Four potentiometers represent cell-voltage inputs. The ESP32 computes battery metrics, supervises a relay output and simulated feedback contact, updates a character LCD, and sends telemetry to Blynk through HTTPS. The design is intended to demonstrate modular monitoring and protection logic; it is not a certified battery pack controller.

Abstract

This project presents the design and simulation of a Smart Multi-Cell Battery Management System (BMS) using an ESP32 DevKit C V4, Wokwi, PlatformIO and Blynk IoT. The system monitors four simulated cell-voltage channels and calculates individual cell percentages, average pack voltage, estimated state of charge (SoC), weakest and strongest cells, voltage imbalance, imbalance trend and an adaptive imbalance threshold. A non-blocking protection subsystem uses debounce, hysteresis, sensor-anomaly detection and timed relay recovery. A supervisory fault state machine classifies faults by source and identifier and manages NORMAL, DEGRADED, FAILSAFE and SHUTDOWN states. A differential LCD engine rotates between battery, system and telemetry pages while giving priority to fault and shutdown screens.

The integrated telemetry subsystem uses a non-blocking Wi-Fi connection manager, a fixed-size 24-entry FIFO event queue and HTTPS batch updates to Blynk. It records boot, state-transition, significant battery-change, Wi-Fi-link and heartbeat events. The analytics layer computes a transparent prototype risk score using imbalance, SoC, trend, recent-fault frequency and supervisory state, and provides rule-based maintenance recommendations. The report summarizes the implementation of all six internship tasks and the available verification evidence. It also identifies simulation limitations, test-build distinctions and the additional hardware and validation needed before any real battery deployment.

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1. Introduction and Project Objectives

1.1 Background

A battery pack is formed by connecting multiple cells in series or parallel to meet a required voltage and capacity. Cell-to-cell variation can cause unequal voltage conditions. Monitoring only the pack average may hide an individual weak or high-voltage cell; therefore, a BMS should retain per-cell measurements and evaluate both individual limits and pack-level imbalance.

This internship project develops a modular monitoring and protection demonstrator. The ESP32 reads four simulated analogue inputs, applies battery-analysis and protection logic, presents local status on a 16 × 2 LCD, and communicates health information to a Blynk dashboard. The design is split into six tasks so that the battery engine, relay protection, LCD, supervisory fault logic, telemetry and analytics can be implemented and verified as cooperating modules.

1.2 Project objectives

- Develop a cell-count-independent battery analysis engine using a compile-time cell-count constant and a shared snapshot interface.
- Identify the weakest and strongest cells, calculate imbalance and trend, and vary the imbalance threshold with estimated SoC.
- Implement non-blocking relay protection with debounce, hysteresis, sensor anomaly detection and timed recovery.
- Develop a differential LCD engine with rotating information pages and higher-priority fault / shutdown displays.
- Build a supervisory state machine that classifies faults and requires staged verification before returning to NORMAL.
- Implement event-triggered cloud telemetry, bounded offline buffering, Wi-Fi retry handling and a Blynk dashboard.
- Add risk scoring, recent-fault tracking, transition history and rule-based maintenance guidance.

1.3 Project scope

The implemented simulation uses four potentiometers as voltage-input substitutes. It demonstrates the software structure for a larger cell count, but it does not physically sense sixteen real cells. The relay and its feedback input are represented by ESP32 GPIO and a slide switch. No electrochemical battery model, real contactor, cell-balancing power stage or isolated high-voltage measurement front end is included in the Wokwi model.

> **Important interpretation**
>
> The percentage displayed as SoC is a linear voltage-normalized estimate used for this demonstrator. It is not a chemistry-specific state-of-charge estimator and should not be used for real battery control.

2. System Requirements and Design Scope

Area	Requirement / design target	Implemented approach
Battery monitoring	Per-cell measurement and pack metrics	Four ADC channels, compile-time array, BMSSnapshot

Area	Requirement / design target	Implemented approach
Adaptive analysis	Weakest / strongest, imbalance, trend and SoC-dependent limit	Single-pass cell analysis and threshold bands
Protection	Non-blocking trip / recovery, noise filtering and sensor diagnostics	Relay FSM, 5-sample history, debounce and hysteresis
Display	Three information pages, low flicker and fault override	16 × 2 I2C LCD with 250 ms differential refresh and 3 s page rotation
Supervision	Source/ID classification and recovery ladder	NORMAL / DEGRADED / FAILSAFE / SHUTDOWN state machine
Telemetry	Event transmission and offline retention	HTTPS batch update, 24-entry FIFO and Wi-Fi manager
Analytics	Trend and fault history, risk indicator and advice	Heuristic risk score and rolling 10-minute fault window
Verification	Demonstrable evidence for all six tasks	Wokwi simulation, Serial Monitor logs and Blynk screenshots

2.1 Functional requirements

- Read and process each enabled cell input on every control cycle.
- Update relay and supervisory state machines without using long blocking delays in the safety path.
- Keep telemetry work on a separate network task so network operations do not directly block the battery and relay logic.
- Buffer telemetry events while Wi-Fi is unavailable and replay from the FIFO when communication is restored.
- Expose battery, relay, supervisory, network and analytics state through Blynk datastreams and Timeline events.

2.2 Non-functional considerations

The implementation emphasizes modularity, bounded memory for event and history buffers, clear separation between control and network activity, deterministic state transitions, and observable logs. The current code is a simulation prototype: it does not provide certified functional safety, redundant sensing, persistent non-volatile event storage, or authenticated TLS certificate validation.

3. System Architecture

The architecture is divided into simulated inputs, ESP32 control functions, local outputs, a network task and the Blynk cloud interface. The control / safety path is executed on Core 1. The Wi-Fi and Blynk HTTP task runs on Core 0. A mutex-protected snapshot is the defined cross-core data hand-off. This reduces coupling between measurement processing and network delivery.

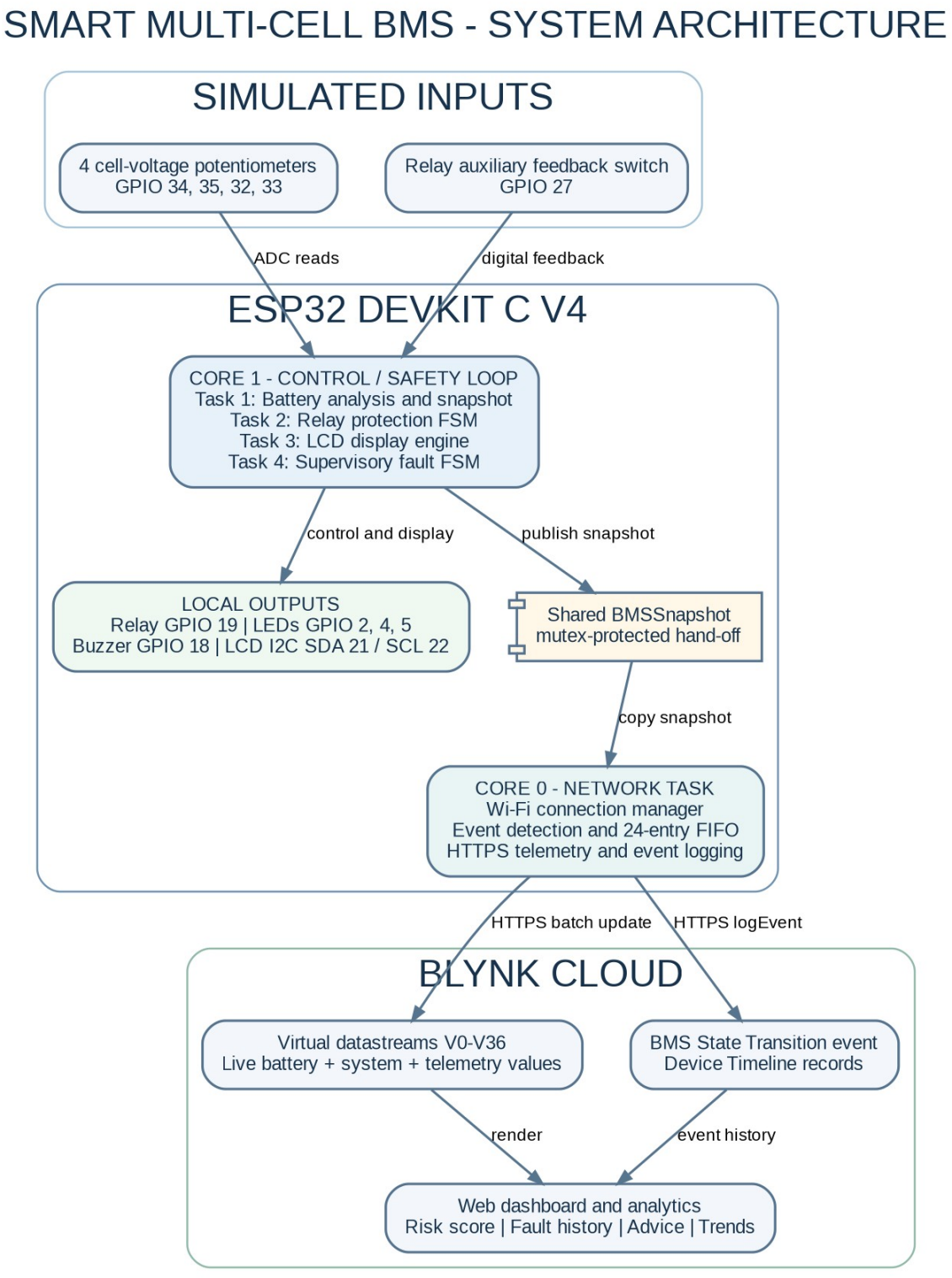


Figure 1. Logical architecture of the Smart Multi-Cell BMS prototype.

3.1 Control and safety path

The control path reads the four analogue inputs and relay feedback, updates the battery snapshot, evaluates the Task 2 relay protection logic, updates the Task 4 supervisory state, and refreshes the LCD and local

indicators. Relay control is handled locally; a loss of cloud connectivity is classified by the supervisory layer but the safety loop is not designed to wait for a cloud request before processing local inputs.

3.2 Network and cloud path

The network task manages Wi-Fi connection states, observes changes in battery and supervisory snapshots, creates telemetry events and attempts HTTPS delivery. State-transition events are also sent to the Blynk Device Timeline using the configured custom event code bms_transition. The Blynk dashboard displays current numeric and text values, while the History-enabled datastreams provide trend charts.

3.3 Data flow and workflow

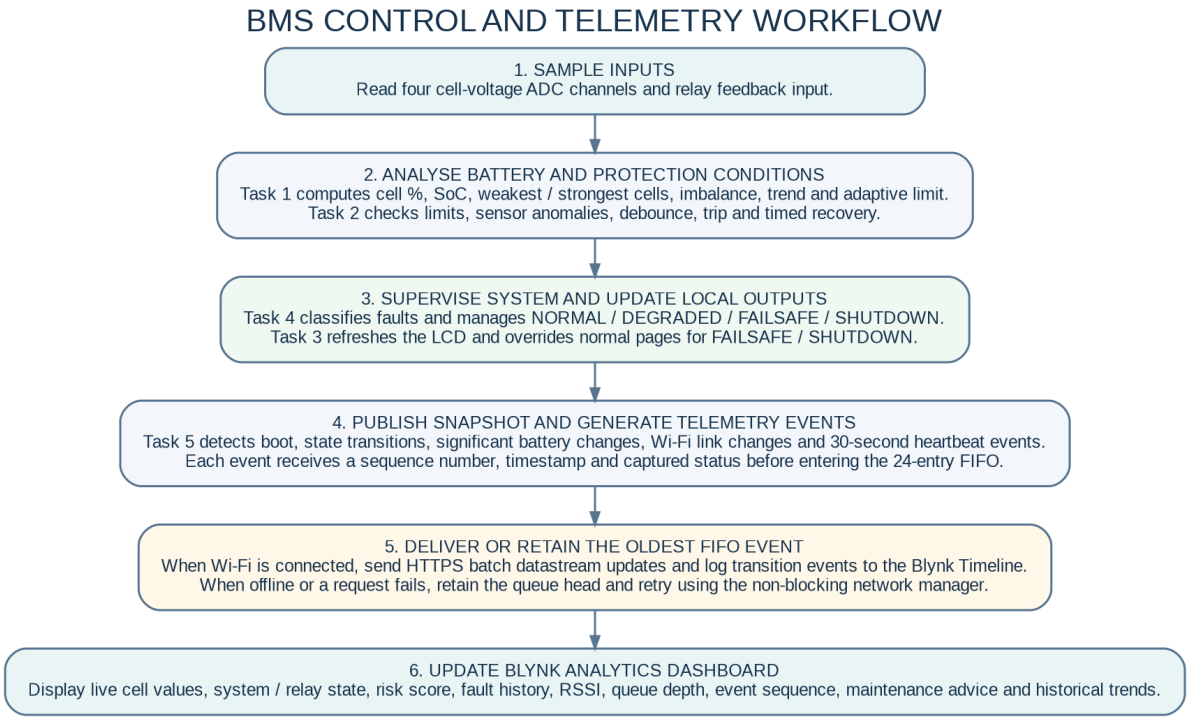


Figure 2. BMS control, fault supervision, event queuing and cloud-delivery workflow.

4. Hardware and Software Implementation

4.1 Hardware represented in Wokwi

Component / signal	ESP32 connection	Role in simulation
Cell 1 potentiometer	GPIO 34	Simulated cell 1 voltage input
Cell 2 potentiometer	GPIO 35	Simulated cell 2 voltage input
Cell 3 potentiometer	GPIO 32	Simulated cell 3 voltage input
Cell 4 potentiometer	GPIO 33	Simulated cell 4 voltage input
LCD I2C	SDA 21, SCL 22; address 0x27	16 × 2 local display
Relay command	GPIO 19	Simulated protection relay output
Relay feedback switch	GPIO 27	Simulated auxiliary feedback contact

Component / signal	ESP32 connection	Role in simulation
Red / green / yellow LEDs	GPIO 2 / 4 / 5	Local status indication
Buzzer	GPIO 18	Audible fault indication

The relay-feedback slide switch represents the auxiliary contact of a relay / contactor. In the documented test arrangement, RIGHT produces a LOW feedback input (healthy) and LEFT produces HIGH (mismatch). The feedback input has a pull-down resistor in the simulation circuit. The physical Wokwi circuit uses four potentiometers rather than actual battery cells.

4.2 Software environment

Tool / platform	Use in the project
ESP32 DevKit C V4	Microcontroller target for the integrated BMS simulation
Arduino framework	Firmware development environment
PlatformIO in Visual Studio Code	Project build and dependency management
Wokwi	Virtual hardware circuit and firmware simulation
Blynk IoT	Cloud datastreams, device dashboard, history and events
LiquidCrystal_I2C	I2C character LCD control
WiFi / HTTPClient / WiFiClientSecure	Wi-Fi connectivity and HTTPS telemetry

4.3 Multi-core organisation

Core 1 owns the live BMSSnapshot and runs battery analysis, protection, supervision, Serial status output and LCD rendering. The snapshot is published to bmsShared using a mutex. Core 0 copies the shared snapshot and runs network management, event generation and HTTPS transactions. The control-side snapshot publication uses a zero-wait mutex acquisition; if the mutex is busy, the telemetry snapshot can remain one cycle behind rather than delaying the safety loop.

5. Task 1 - Modular Battery Management Engine

5.1 Design and data structures

The cell count is defined by the compile-time constant NUM_CELLS. The current build sets NUM_CELLS = 4 and stores cell values in BatteryCell cells[NUM_CELLS]. Each BatteryCell contains a floating-point voltage and an integer percentage. BMSSnapshot acts as the public data structure for all calculated battery measurements and summary values. The runBatteryEngine() function updates the snapshot, while getSnapshot() exposes it to the other control modules.

```
constexpr uint8_t NUM_CELLS = 4;
struct BatteryCell { float voltage; int percent; };
struct BMSSnapshot {
  uint8_t cellCount;
  BatteryCell cells[NUM_CELLS];
  float averageVoltage, batterySoC;
  uint8_t weakestCell, strongestCell;
```

```
  float minimumVoltage, maximumVoltage, voltageImbalance;
  char imbalanceTrend[8];
  float adaptiveThreshold;
  char status[12];
};
```

5.2 Measurement and analysis

Each ESP32 ADC input is read as a 12-bit value. The prototype converts the raw code to an input voltage using a 0–3.3 V range. The per-cell percentage is then calculated by linearly normalising that voltage to 0–100%. Average voltage and the normalised pack SoC are calculated from the four-cell average. This mapping is used to make potentiometer movement easy to interpret during simulation; a real battery would require cell-chemistry-aware voltage limits and state estimation.

```
Vcell = (ADCraw / 4095.0) × 3.3
Cell percentage = constrain((Vcell / 3.3) × 100, 0, 100)
Average voltage = sum(Vcell[i]) / NUM_CELLS
Prototype SoC = (Average voltage / 3.3) × 100
Voltage imbalance = Vmaximum - Vminimum
```

A single loop finds the minimum and maximum voltages and records the corresponding weakest and strongest cell indices. Imbalance trend is obtained by comparing the present imbalance with the previous value. A tolerance of 0.005 V is used to classify the trend as RISING, FALLING or STABLE.

5.3 Adaptive imbalance threshold

Estimated SoC band	Adaptive imbalance threshold
SoC ≥ 80%	0.05 V
50% ≤ SoC < 80%	0.08 V
20% ≤ SoC < 50%	0.10 V
SoC < 20%	0.12 V

The battery status prioritises weak-cell detection and high-voltage detection before imbalance. If neither individual-cell condition is active, the engine compares the pack imbalance to the adaptive threshold and reports IMBALANCE or NORMAL.

Adaptive-threshold scope limitation

The final implementation selects the adaptive imbalance limit from the estimated SoC bands. A separate discharge-rate input is not modelled because the current Wokwi circuit has no current sensor or discharge-load model; discharge-rate compensation is therefore not claimed as implemented.

```
================================================
        MODULAR BMS ENGINE - TASK 1
================================================
Number of Cells : 4

Cell 1 : 1.358 V | 41 %
Cell 2 : 1.687 V | 51 %
Cell 3 : 1.894 V | 57 %
Cell 4 : 1.516 V | 45 %
------------------------------------------------
Average Voltage : 1.614 V
Battery SoC     : 48.9 %
Weakest Cell    : Cell 1 (1.358 V)
Strongest Cell  : Cell 3 (1.894 V)
Voltage Imbal.  : 0.536 V
Imbalance Trend  : STABLE
Adaptive Limit  : 0.100 V
System Status   : IMBALANCE
Blynk Status    : CONNECTED
```

Figure 3. Task 1 cell-extreme test: weakest / strongest identification and calculated imbalance.

```
        MODULAR BMS ENGINE - TASK 1
================================================
Number of Cells : 4

Cell 1 : 1.065 V | 32 %
Cell 2 : 1.113 V | 33 %
Cell 3 : 1.100 V | 33 %
Cell 4 : 1.087 V | 32 %
------------------------------------------------
Average Voltage : 1.091 V
Battery SoC     : 33.1 %
Weakest Cell    : Cell 1 (1.065 V)
Strongest Cell  : Cell 2 (1.113 V)
Voltage Imbal.  : 0.048 V
Imbalance Trend  : STABLE
Adaptive Limit  : 0.100 V
System Status   : NORMAL
Blynk Status    : CONNECTED
================================================
```

Figure 4. Adaptive threshold verification at an estimated SoC of 33.1%.

5.4 Scalability analysis

The processing loops are parameterised by NUM_CELLS. Increasing the compile-time count from four to sixteen cells therefore reuses the same analysis functions and increases the cell-processing work

approximately linearly. With an assumed BatteryCell size of 8 bytes, the cell-data array occupies approximately 32 bytes for four cells and 128 bytes for sixteen cells. This estimate covers only the BatteryCell array and excludes ADC history, snapshots, event buffers, stack and other firmware state.

Parameter	4-cell configuration	16-cell software configuration
Cell entries	4	16
BatteryCell data (assumed 8 bytes each)	32 bytes	128 bytes
Cell data growth	Baseline	4×
Analysis time complexity	O(N)	O(N)
Core analysis logic	Same loops	Same loops

Hardware scalability limitation

The software data model scales to sixteen cells, but the current ESP32 simulation wiring does not provide sixteen physical cell inputs. A real larger pack would need a suitable external ADC, analogue multiplexer and safe measurement front end with appropriate isolation / scaling.

5.5 Task 1 verification summary

Test	Test objective	Observed result
A	Weakest / strongest cell and imbalance	PASS - Cell 1 and Cell 3 correctly identified; 0.536 V imbalance calculated.
B	Rising and falling imbalance trend	PASS - trend changed according to the direction of the cell-voltage spread.
C	SoC-dependent threshold	PASS - 0.100 V at 33.1% SoC and 0.080 V at 68.1% SoC.
D	4-to-16 software scalability	Architecture analysis completed; physical 16-cell test not performed.
E	Blynk dashboard integration	PASS - battery metrics displayed on the dashboard.

6. Task 2 - Non-Blocking Protection Relay and Safety System

6.1 Relay state machine

The protection relay is managed by a four-state finite-state machine: CONNECTED, TRIP_PENDING, TRIPPED and RECOVERING. A fault condition does not cause an immediate trip. The system first enters TRIP_PENDING and requires the fault to remain present for the debounce interval. If the fault clears before the timer expires, the system returns to CONNECTED. After a trip, the relay remains TRIPPED until the fault is cleared and the timed recovery conditions are satisfied.

State	Purpose / transition behaviour
CONNECTED	Normal relay operation; protection conditions continuously evaluated.
TRIP_PENDING	A candidate fault is present; a 400 ms debounce period filters short transients.
TRIPPED	Protection output is in the safe / tripped condition while the fault persists.
RECOVERING	Fault has cleared; a 3 s recovery interval and hysteresis conditions prevent immediate re-energisation.

6.2 Protection thresholds and anomaly checks

Protection item	Configured behaviour in the final source
Weak-cell trip / release	Trip below 30%; release after recovery above 35%.
High-cell trip / release	Trip above 90%; release below 85%.
Debounce	400 ms persistence before confirmed relay trip.
Timed recovery	3,000 ms after fault clearance, with recovery conditions.
Sample window	Five stored samples per cell for statistical checks.
Out-of-range input	Values below 0.02 V or above 3.28 V are treated as invalid.
Abrupt voltage jump	Absolute minimum 0.30 V; statistical outlier comparison uses 6× standard deviation with a 0.02 V floor.
Frozen input	Cell activity is assessed over a 4,000 ms observation window while pack activity is considered.

The detector uses a per-cell history and activity statistics to distinguish a short disturbance from a persistent abnormal condition. The frozen-sensor check is based on sequential activity evidence rather than assuming that all cells must change simultaneously. This matters when the simulated potentiometers are adjusted at different times.

```
==============================================
  MODULAR BMS ENGINE - TASK 1 + TASK 2 SAFETY v3
==============================================
Number of Cells : 4

Cell 1 : 1.700 V | 51 %
Cell 2 : 1.710 V | 51 %
Cell 3 : 1.710 V | 51 %
Cell 4 : 1.700 V | 51 %
----------------------------------------------
Average Voltage : 1.705 V
Battery SoC     : 51.7 %
Voltage Imbal.  : 0.010 V
Imbalance Trend : STABLE
Adaptive Limit  : 0.080 V
Battery Status  : NORMAL
----------------------------------------------
Relay State     : RECOVERING
Fault Reason    : NONE
Wi-Fi Status    : CONNECTED
Blynk Status    : CONNECTED
==============================================
[RELAY LOG] t=51574ms | RECOVERING -> CONNECTED | reason: recovery period complete
[RELAY LOG] t=51696ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (1.271V, window mean 1.652V)
[RELAY LOG] t=51697ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=51729ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (0.990V, window mean 1.271V)
[RELAY LOG] t=51730ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=51762ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (0.781V, window mean 0.990V)
[RELAY LOG] t=52162ms | TRIP_PENDING -> TRIPPED | reason: Cell 1 WEAK (13%)
```

Figure 5. Weak-cell protection and debounce-related relay state evidence.

```
==============================================
  MODULAR BMS ENGINE - TASK 1 + TASK 2 SAFETY v3
==============================================
Number of Cells : 4

Cell 1 : 1.736 V | 52 %
Cell 2 : 1.710 V | 51 %
Cell 3 : 1.710 V | 51 %
Cell 4 : 1.700 V | 51 %
----------------------------------------------
Average Voltage : 1.714 V
Battery SoC     : 51.9 %
Voltage Imbal.  : 0.035 V
Imbalance Trend : STABLE
Adaptive Limit  : 0.080 V
Battery Status  : NORMAL
----------------------------------------------
Relay State     : CONNECTED
Fault Reason    : NONE
Wi-Fi Status    : CONNECTED
Blynk Status    : CONNECTED
==============================================
[RELAY LOG] t=81729ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (1.126V, window mean 1.736V)
[RELAY LOG] t=81730ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=81762ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (0.794V, window mean 1.126V)
[RELAY LOG] t=82162ms | TRIP_PENDING -> TRIPPED | reason: Cell 1 WEAK (4%)
```

Figure 6. Transient disturbance rejected before the debounce interval expires.

```
================================================
  MODULAR BMS ENGINE - TASK 1 + TASK 2 SAFETY v3
================================================
Number of Cells : 4

Cell 1 : 1.736 V | 52 %
Cell 2 : 1.710 V | 51 %
Cell 3 : 1.710 V | 51 %
Cell 4 : 1.700 V | 51 %
------------------------------------------------
Average Voltage : 1.714 V
Battery SoC     : 51.9 %
Voltage Imbal.  : 0.035 V
Imbalance Trend : STABLE
Adaptive Limit  : 0.080 V
Battery Status  : NORMAL
------------------------------------------------
Relay State     : CONNECTED
Fault Reason    : NONE
Wi-Fi Status    : CONNECTED
Blynk Status    : CONNECTED
================================================
[RELAY LOG] t=318936ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (1.942V, window mean 1.736V)
[RELAY LOG] t=318937ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=318969ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (2.103V, window mean 1.942V)
[RELAY LOG] t=318970ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=319002ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (2.310V, window mean 2.103V)
[RELAY LOG] t=319004ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=319070ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (3.252V, window mean 2.310V)
[RELAY LOG] t=319270ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=319303ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (2.248V, window mean 2.603V)
[RELAY LOG] t=319304ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=319336ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (2.042V, window mean 2.248V)
[RELAY LOG] t=319337ms | TRIP_PENDING -> CONNECTED | reason: condition cleared before debounce (filtered as noise)
[RELAY LOG] t=319402ms | CONNECTED -> TRIP_PENDING | reason: Cell 1 JUMP (1.652V, window mean 2.042V)
```

Figure 7. Rapid voltage-jump detection and transient rejection.

```
================================================
  MODULAR BMS ENGINE - TASK 1 + TASK 2 SAFETY v3
================================================
Number of Cells : 4

Cell 1 : 1.662 V | 50 %
Cell 2 : 1.653 V | 50 %
Cell 3 : 0.980 V | 29 %
Cell 4 : 1.771 V | 53 %
------------------------------------------------
Average Voltage : 1.516 V
Battery SoC     : 46.0 %
Voltage Imbal.  : 0.791 V
Imbalance Trend : FALLING
Adaptive Limit  : 0.100 V
Battery Status  : WEAK CELL
------------------------------------------------
Relay State     : TRIPPED
Fault Reason    : Cell 1 FROZEN (pack active, this cell static)
Wi-Fi Status    : CONNECTED
Blynk Status    : CONNECTED
================================================
------------------------------------------------
[TASK 2J] Controlled test stimulus complete.
[TASK 2J] Returning to live ADC values.
================================================
[RELAY LOG] t=11013ms | TRIPPED -> RECOVERING | reason: fault cleared, starting timed recovery
```

Figure 8. Frozen-sensor detection while the remaining pack inputs remain active.

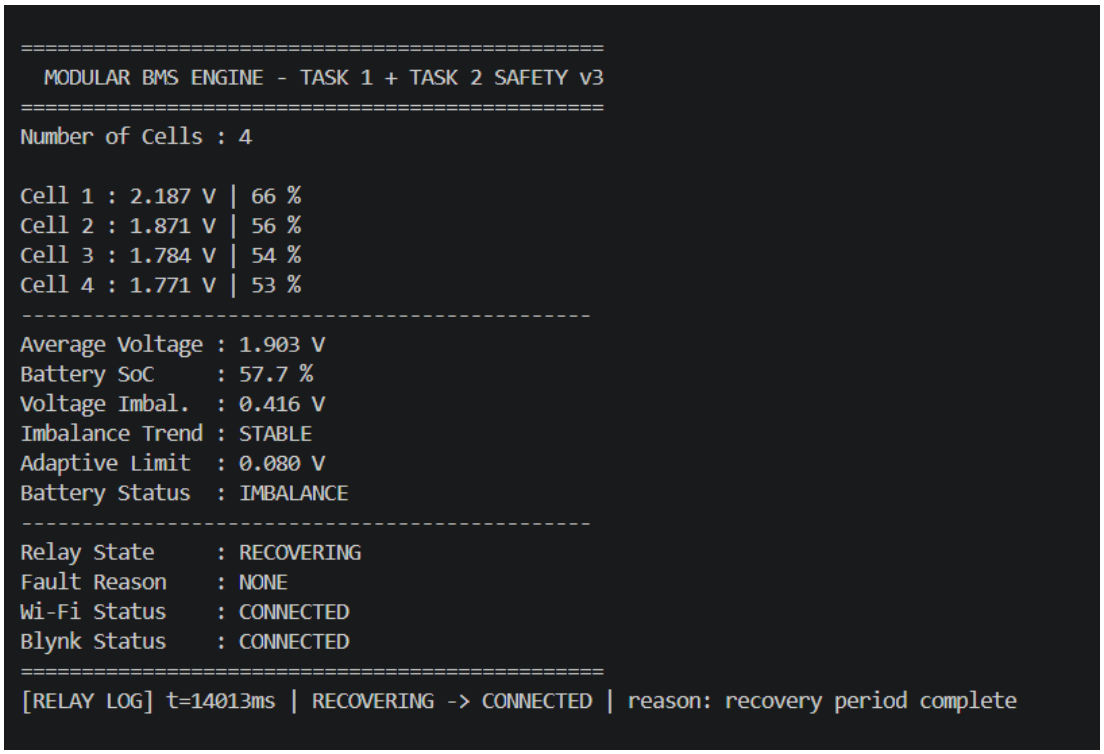


Figure 9. Completion of timed recovery after sensor-fault clearance.

6.3 Task 2 verification summary

Test	Scenario	Result
2A	All cells in normal operating range	PASS - normal status and relay remained connected.
2B	Cell below weak-cell threshold	PASS - TRIP_PENDING followed by TRIPPED after debounce.
2C	Short transient / noise disturbance	PASS - condition cleared before debounce; no confirmed trip.
2D / 2F	Fault clearance and timed recovery	PASS - recovery interval completed before CONNECTED.
2E	High-voltage cell	PASS - high threshold trip and hysteresis release verified.
2G	Fault reappears during recovery	PASS - returned to TRIPPED and recovery was cancelled.
2H	Out-of-range sensor input	PASS - 0.000 V input detected and protected.
2I	Unrealistic rapid voltage jump	PASS - short jump event filtered when it cleared before debounce.
2J	Frozen sensor	PASS - constant cell detected while other inputs were active.
2K	Repeated fault injection	PASS - at least seven cycles completed without observed reset / stuck state.

7. Task 3 - Flicker-Free LCD Display Engine

7.1 Display pages and timing

The LCD engine controls a 16 × 2 I2C display at address 0x27. During normal operation, the display rotates through three pages: Battery, System and Telemetry. Each page remains visible for 3 seconds. The content refresh timer is 250 ms (4 Hz), allowing values to update without using long blocking delays in the control loop.

Page	Example information
Battery	Battery status, estimated SoC and adaptive threshold.
System	Relay state, supervisory state, weakest / strongest cell and Blynk link indicator.
Telemetry	Average voltage, voltage imbalance, imbalance trend and Wi-Fi indicator.
FAILSAFE override	Dedicated fault display with source and short fault identifier.
SHUTDOWN override	Dedicated shutdown display and fault identifier.

7.2 Differential rendering

The LCD engine stores the characters currently shown in a two-row screen buffer. At each refresh, the new page strings are compared against the previous rendered characters. Only positions whose characters changed are written to the LCD. This avoids clearing the whole display on every update and reduces visible flicker.

When the supervisory state is FAILSAFE or SHUTDOWN, the normal page rotation is suspended and the corresponding fault screen takes priority. Once recovery succeeds and the system returns to NORMAL or DEGRADED, normal page rotation resumes.

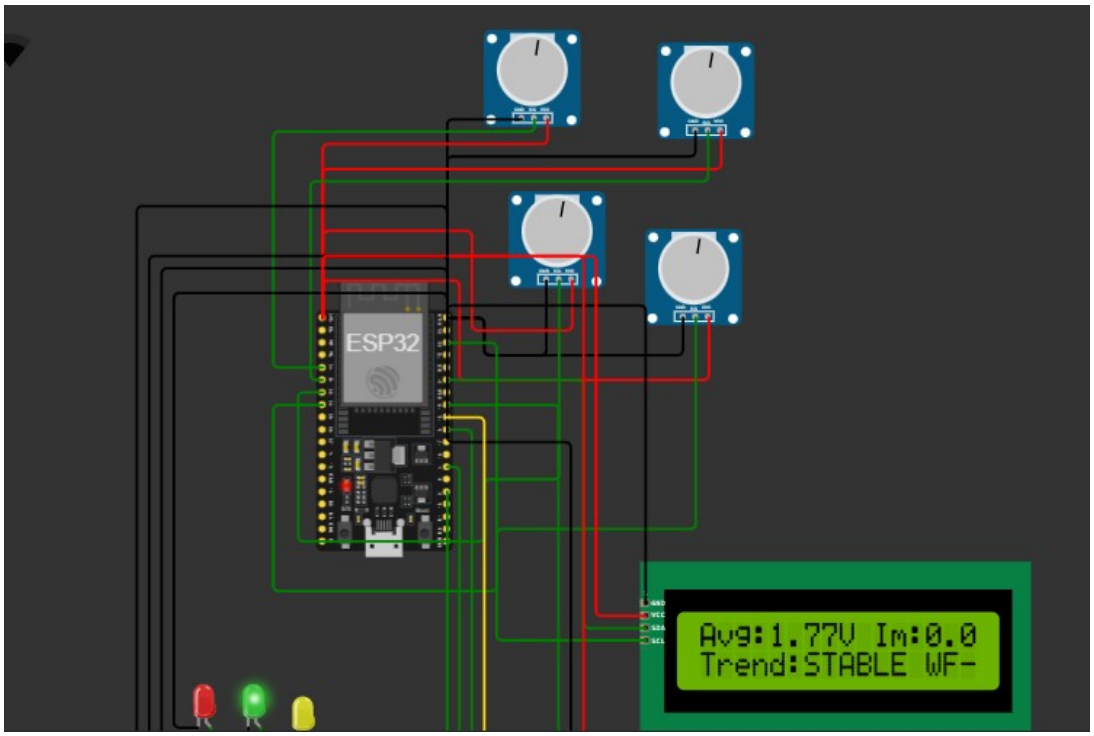


Figure 10. Wokwi circuit and LCD information page during normal operation.

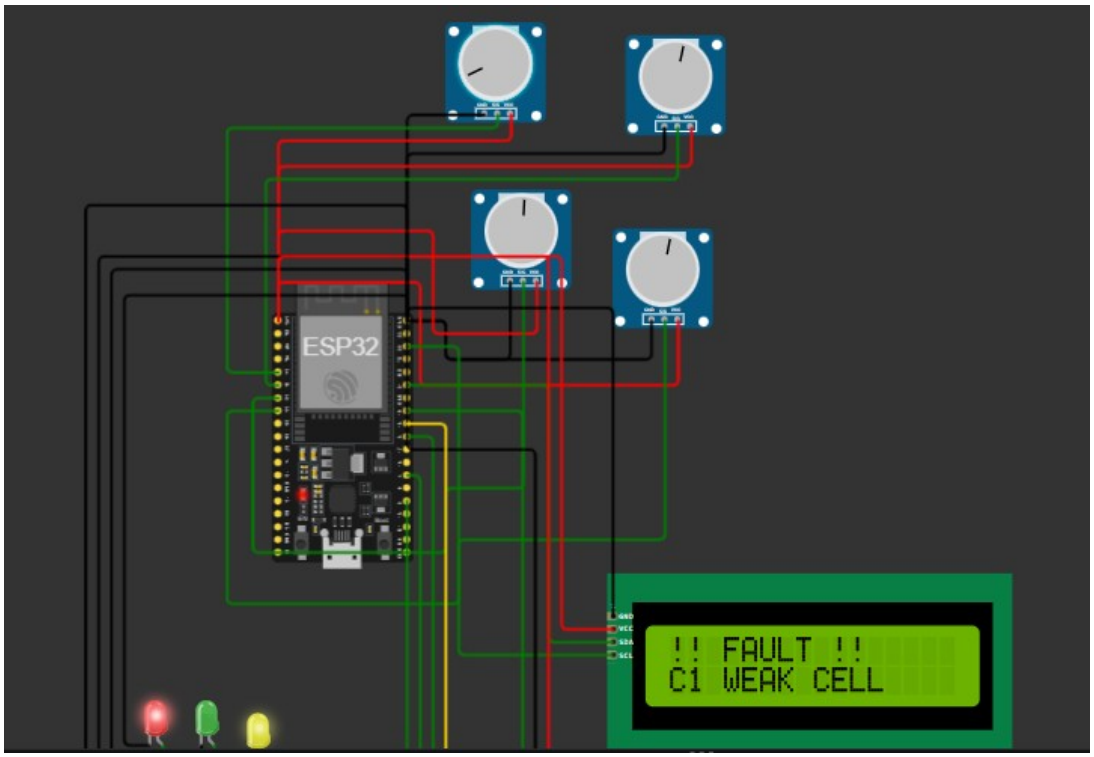


Figure 11. Critical fault override replacing the normal LCD information page.

7.3 Task 3 verification summary

Test	Verification	Result
3B	Automatic three-page rotation	PASS - Battery, System and Telemetry pages rotate every 3 s.

Test	Verification	Result
3C	Non-blocking display updates	PASS - measurements and safety logic continued during potentiometer adjustment.
3D	Differential rendering	PASS - changed characters updated without full-screen flicker.
3E	Critical fault override	PASS - dedicated fault screen displayed on critical fault.
3F / 3G	Fault-screen persistence and override priority	PASS - normal rotation remained suspended while fault persisted.
3H	Recovery and return to display rotation	PASS - normal display resumed after protection recovery.

8. Task 4 - Fault State Machine and Structured Recovery

8.1 Supervisory states

State	Meaning and action
NORMAL	No active supervisory fault. Normal battery monitoring, relay supervision and telemetry continue.
DEGRADED	A non-critical condition such as sustained communication loss is present; system continues under degraded status.
FAILSAFE	A critical battery, ADC or relay fault is detected. Fault identity is retained and protective action is applied.
SHUTDOWN	Latched terminal protective state. The relay is forced to its safe / tripped state; restart is required to leave SHUTDOWN.

8.2 Fault classification

Fault source	Representative identifiers / conditions
BATTERY	BATTERY_WEAK and BATTERY_HIGH based on cell percentage limits.
ADC	ADC_FROZEN, ADC_OUT_OF_RANGE and abnormal jump / measurement conditions.
RELAY	RELAY_MISMATCH when relay command and feedback disagree persistently.
COMMUNICATION	COMMUNICATION when Wi-Fi / Blynk connectivity remains unavailable beyond the communication-loss threshold.

8.3 Timing and recovery ladder

In the final integrated source, relay-feedback mismatch is ignored for a 2-second boot-settling interval and must persist for 500 ms before it is classified. A communication loss is evaluated over a 10-second threshold. FAILSAFE escalates to SHUTDOWN if it remains unresolved for 5 seconds. Before returning from FAILSAFE to NORMAL, the system performs a sequential recovery ladder: ADC, BATTERY, RELAY and COMMUNICATION. Each stage must remain clean for 1 second. Fault source and ID are retained during the non-normal state and are cleared only after successful recovery.

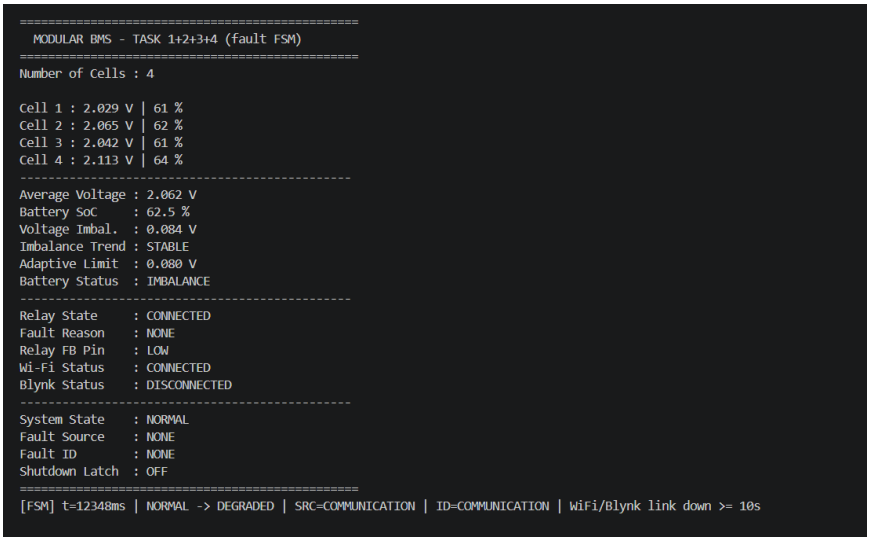


Figure 12. Earlier communication-fault isolation test using the temporary SW2 verification fixture.

The earlier Task 4 communication-fault test used a temporary slide switch SW2 on GPIO26 to force the Blynk link status to DISCONNECTED while the battery and relay remained healthy. This fixture was used because Wokwi made repeatable network interruption difficult. SW2 was removed from the final circuit and is not part of the final hardware architecture. The final firmware instead monitors the actual Wi-Fi and Blynk HTTP status.

```
================================================
[FSM] Recovery check passed: COMMUNICATION
[FSM] t=63675ms | FAILSAFE -> NORMAL | SRC=BATTERY | ID=BATTERY_WEAK | RECOVERY=PASS

================================================
  MODULAR BMS - TASK 1+2+3+4 (fault FSM)
================================================
Number of Cells : 4

Cell 1 : 2.016 V | 61 %
Cell 2 : 1.955 V | 59 %
Cell 3 : 2.090 V | 63 %
Cell 4 : 2.029 V | 61 %
------------------------------------------------
Average Voltage : 2.023 V
Battery SoC     : 61.3 %
Voltage Imbal.  : 0.135 V
Imbalance Trend : STABLE
Adaptive Limit  : 0.080 V
Battery Status  : IMBALANCE
------------------------------------------------
Relay State     : CONNECTED
Fault Reason    : NONE
Relay FB Pin    : LOW
Wi-Fi Status    : CONNECTED
Blynk Status    : CONNECTED
------------------------------------------------
System State    : NORMAL
Fault Source    : NONE
Fault ID        : NONE
================================================
```

Figure 13. Earlier Task 4 evidence showing recovery checks and return to NORMAL after fault clearance.

8.4 Final integrated relay-mismatch recovery test

A later final-firmware test injected relay mismatch through the GPIO27 feedback switch. The supervisory FSM changed from NORMAL to FAILSAFE and retained the source RELAY and identifier RELAY_MISMATCH. The recovery sequence then passed ADC, BATTERY, RELAY and COMMUNICATION checks. The timestamped log showed NORMAL to FAILSAFE at 9,022 ms and FAILSAFE to NORMAL at 13,047 ms, giving an observed total recovery interval of approximately 4.025 seconds. Both transitions were logged to Blynk.

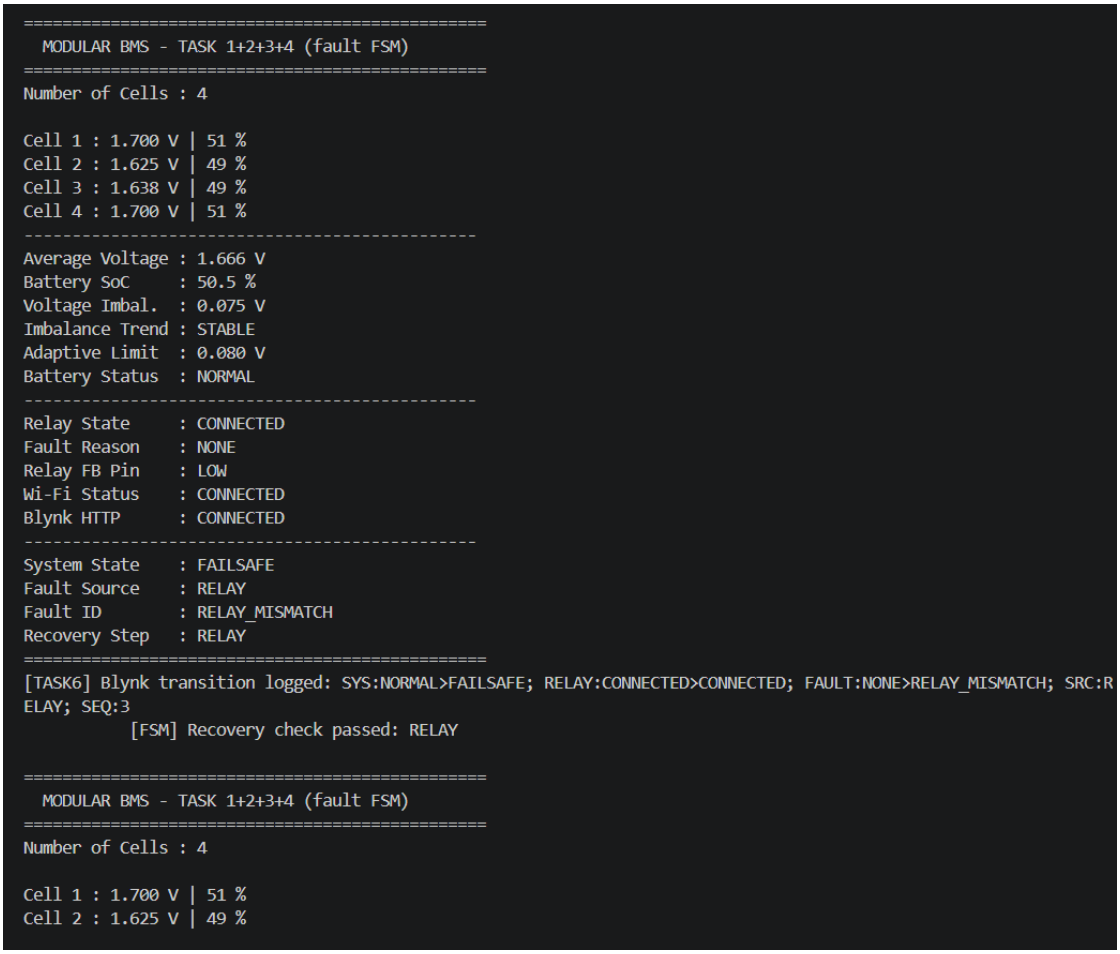


Figure 14. Final integrated firmware: relay-mismatch transition recorded by the Blynk transition-event logger.

```
[FSM] t=9022ms | NORMAL -> FAILSAFE | SRC=RELAY | ID=RELAY_MISMATCH
[FSM] Recovery check passed: ADC
[FSM] Recovery check passed: BATTERY
[FSM] Recovery check passed: RELAY
[FSM] Recovery check passed: COMMUNICATION
[FSM] t=13047ms | FAILSAFE -> NORMAL | SRC=RELAY | ID=RELAY_MISMATCH | RECOVERY=PASS
[TASK6] Blynk transition logged: SYS:FAILSAFE>NORMAL ... SEQ:4
```

8.5 Task 4 verification summary

Scenario	Observed result	Evidence / qualification
Communication fault isolation	NORMAL → DEGRADED after 10 s communication loss.	Earlier controlled SW2/GPIO26 test fixture; not part of final circuit.
Relay command-feedback mismatch	NORMAL → FAILSAFE with RELAY / RELAY_MISMATCH.	Final GPIO27 feedback-switch test.
Battery weak-cell isolation	Battery source and BATTERY_WEAK retained during FAILSAFE.	Task 4 uploaded report evidence.
Staged recovery	ADC → BATTERY → RELAY → COMMUNICATION, then FAILSAFE → NORMAL.	Final integrated firmware log; 4.025 s observed elapsed time.

Scenario	Observed result	Evidence / qualification
FAILSAFE timeout	Unresolved fault can escalate to SHUTDOWN after 5 s.	Earlier final dashboard / Serial Monitor test showed this configured behaviour.
SHUTDOWN latch	Relay forced to TRIPPED; restart required to leave the state.	Final firmware design and observed shutdown test.

9. Task 5 - Event-Driven Telemetry and Live Blynk Dashboard

9.1 Telemetry architecture

The real-time Blynk library is not used in the final integrated firmware. Instead, the network task sends data through Blynk Cloud HTTPS endpoints. This approach separates cloud data transmission from the local safety-control path. The network task runs on Core 0 while the battery, relay and system state logic run on Core 1.

9.2 Event generation

Event reason	Trigger
BOOT	Initial valid snapshot after startup.
FSM_TRANSITION	Change in relay or supervisory state; also used for Blynk Timeline logging.
BATTERY_CHANGE	Significant change: cell voltage ≥ 0.05 V, SoC ≥ 3 percentage points or imbalance ≥ 0.03 V.
HEARTBEAT	30-second health update for uptime and historical charts.
WIFI_LINK_CHANGE	Wi-Fi link loss or restoration.

9.3 Offline queue and network manager

Telemetry events are stored in a fixed-size 24-entry FIFO. Each entry captures the battery and supervisory snapshot, event sequence number, event time, fault counters, recent-fault count, network state and RSSI. When the queue becomes full, the oldest event is discarded to preserve the newest system state and the dropped-event counter is incremented. Queue storage is RAM-resident and therefore does not survive ESP32 reset or power loss.

The Wi-Fi manager uses CONNECTING, CONNECTED and RETRY_WAIT states. A connection attempt has a 10-second timeout and the retry interval is 5 seconds. While Wi-Fi is unavailable, events remain queued. After link restoration, the network task attempts the oldest event first. The minimum send gap is 500 ms when the last Blynk transaction succeeded; failed transmission uses a 3-second retry gap. The oldest event is removed only after the required request or requests succeed.

9.4 Blynk interface and event logging

The batch-update endpoint updates the battery and Task 5 / Task 6 datastreams in one request. For state transitions, the firmware also calls the Blynk Event logging endpoint using the custom event code bms_transition. The event description contains the previous and new system state, previous and new relay state, previous and new fault ID, fault source and event sequence. The template event must exist and Timeline

visibility must be enabled. The integration issue encountered during configuration was an HTTP 400 response because the event had not yet been created in the template; after creating BMS State Transition with event code bms_transition and enabling device-page visibility, the Serial Monitor showed successful transition logging.

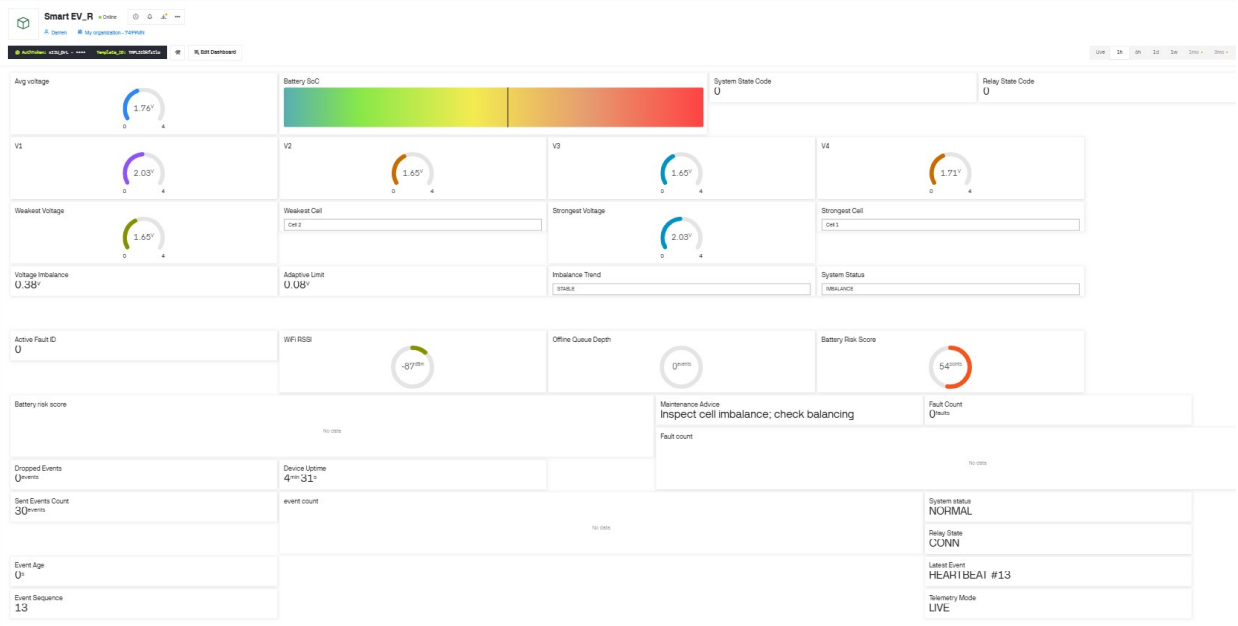


Figure 15. Blynk live dashboard showing battery values, connectivity, queue depth, risk score and maintenance advice.

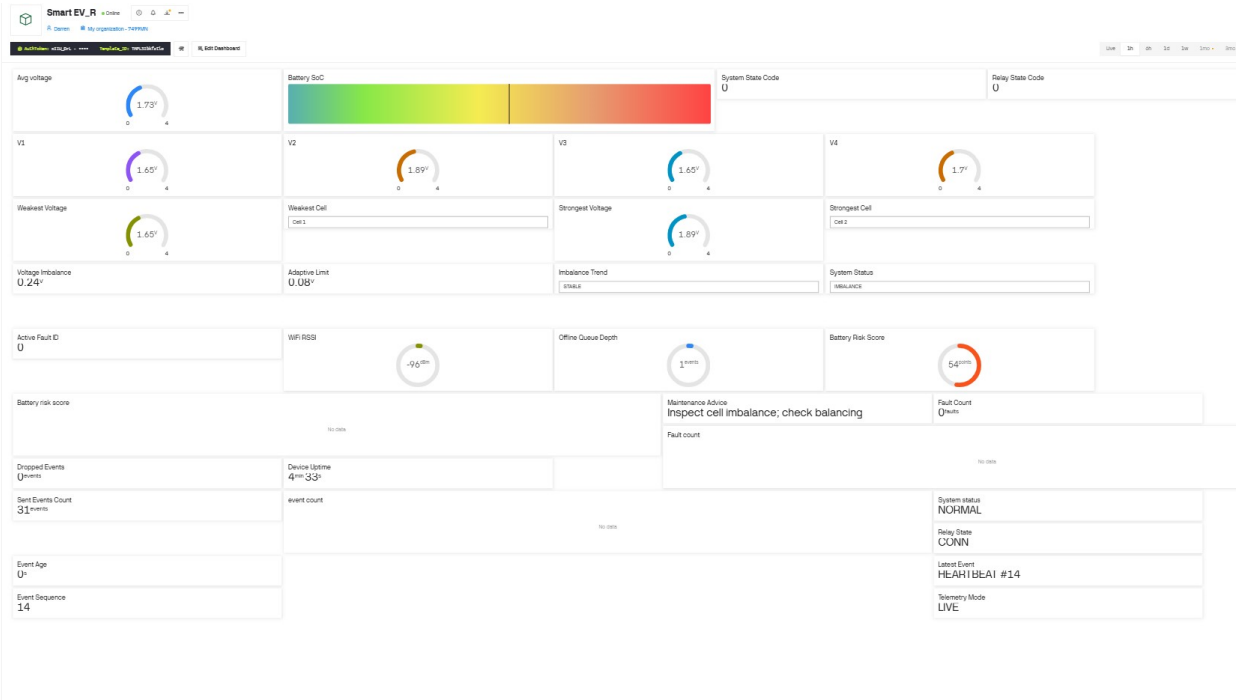


Figure 16. Dashboard snapshot during event accumulation; one event was temporarily present in the queue while the device remained online.

9.5 Verification record and qualification

Live telemetry was observed on the Blynk dashboard. The screenshots show the device online, Wi-Fi RSSI, queue depth, sent-event count, latest event and telemetry mode. The final relay-mismatch test also logged both supervisory transitions to the Blynk Timeline. The significant-parameter-change test was previously marked

inconclusive because the threshold-crossing event was not consistently observed during the slow manual adjustment used in that test; it is therefore not reported as a fully passed test.

The Wi-Fi outage and FIFO replay demonstration was performed earlier with a temporary test build that exposed software controls to force an outage and restore the link. That test showed events accumulating while the simulated link was unavailable and replaying in increasing sequence order after restoration. The forced-outage controls were removed from the final production firmware. The final build contains the actual Wi-Fi reconnect state machine and queue-replay logic; a repeatable final-build physical access-point outage was not separately documented in the supplied evidence.

Evidence wording for evaluation

The report distinguishes (1) implementation present in the final firmware, (2) behavior observed on the final integrated build, and (3) a queue/replay test demonstrated with a temporary verification build. This avoids implying that temporary test controls are part of the final product.

10. Task 6 - Enterprise Analytics and Decision Dashboard

10.1 Risk score model

A composite risk score from 0 to 100 is calculated for each captured telemetry event. It is a transparent prototype heuristic intended to combine battery condition and supervisory severity in one dashboard indicator. It is not a validated state-of-health estimate, remaining-useful-life model or safety certification metric.

Risk component	Maximum contribution	Implemented rule
Voltage imbalance	30 points	Voltage imbalance relative to the current adaptive threshold, clamped to 0–30.
SoC	15 points	(100 − SoC) × 0.15, clamped to 0–15.
Imbalance trend	15 points	RISING = 15; START = 5; STABLE / FALLING = 0.
Recent fault frequency	15 points	5 points per distinct fault in the rolling ten-minute window, capped at 15.
System state	25 points	NORMAL = 0; DEGRADED = 8; FAILSAFE = 20; SHUTDOWN = 25.

The recent-fault window uses a fixed 16-entry timestamp buffer. A distinct new non-NONE fault source / ID increments the fault count. Escalation of the same fault from FAILSAFE to SHUTDOWN does not count as another fault occurrence. This correction is reflected in the dashboard evidence where one relay-mismatch occurrence is displayed as one fault.

10.2 Maintenance recommendations

The maintenance advice is rule-based and selects the first matching condition in a priority sequence. SHUTDOWN prompts inspection before reset; FAILSAFE prompts isolation and inspection; a communication fault suggests checking Wi-Fi and Blynk; repeated recent faults suggest inspection of the pack and protection

hardware; rising imbalance prompts cell-balance inspection; excessive imbalance prompts an imbalance check; low SoC prompts recharge; and higher residual risk values prompt increased monitoring or inspection.

10.3 Dashboard state severity

System state code (V14)	State	Suggested severity
0	NORMAL	Green
1	DEGRADED	Yellow
2	FAILSAFE	Orange
3	SHUTDOWN	Red

Text values are also published so that the dashboard can display readable state names rather than relying only on numeric codes. Relay state is published both numerically and as a short text label. The dashboard provides latest event, event sequence, event age, telemetry mode, RSSI quality and recent-fault count alongside battery measurements.

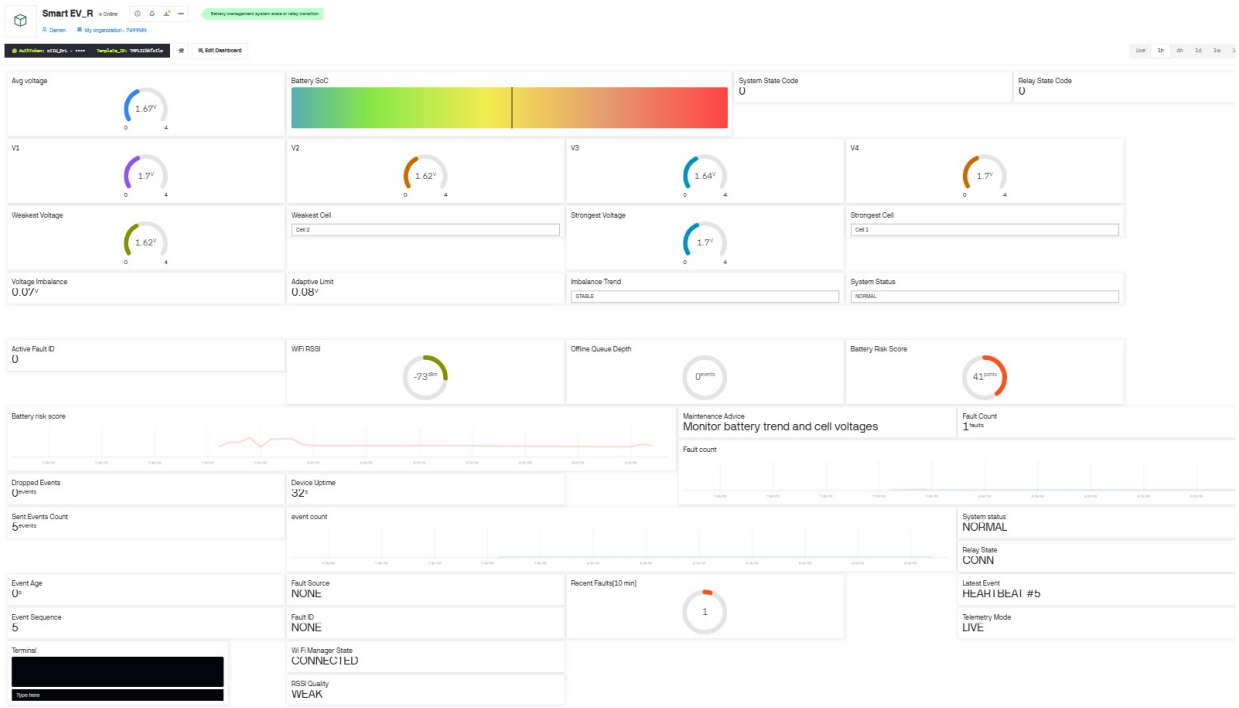


Figure 17. Integrated Blynk analytics dashboard after recovery: NORMAL system state, zero queued events, one recorded fault, one recent fault and visible history traces.

10.4 Dashboard verification summary

Feature	Evidence / result
Current battery and state telemetry	Dashboard displays average voltage, V1–V4, SoC, weakest / strongest cell, imbalance, threshold and system status.
Risk score and maintenance advice	Risk score and textual advice are displayed; example screenshot shows risk score 41 and trend-monitoring advice.

Feature	Evidence / result
Fault frequency	Dashboard displays Fault Count = 1 and Recent Faults (10 min) = 1 for the observed relay-mismatch occurrence.
Queue and telemetry mode	Queue depth returned to 0 and Telemetry Mode displayed LIVE in the latest normal-state screenshot.
Historical charts	Risk-score and fault-count charts show plotted history in the latest dashboard capture; chart population depends on incoming samples and selected History aggregation.
Transition logging	Serial Monitor confirms successful Blynk transition logging; device-page event visibility is enabled in the Blynk template.

11. Verification Methodology and Results

Verification was carried out primarily in Wokwi by adjusting cell-voltage potentiometers, changing the relay-feedback switch and observing the Serial Monitor, LCD and Blynk dashboard. Tests were grouped by task so that each functional requirement had an observable expected result. The uploaded Task 1–4 PDFs contain the original screenshots and written test descriptions; the newer dashboard captures and Serial Monitor output supplement those results for the final Task 5–6 integration.

11.1 Consolidated verification matrix

Task	Verification areas	Evidence status
1	Cell extremes; rising / falling trend; adaptive SoC threshold; 4-to-16 software scalability; dashboard values.	Tests A–C and E documented as PASS; D is software architecture analysis only.
2	Normal operation; weak / high cell trip; debounce; hysteresis; timed recovery; sensor anomalies; repeated fault cycles.	Tests 2A–2K documented as PASS in uploaded Task 2 report.
3	Three-page rotation; non-blocking refresh; differential rendering; fault override; fault persistence; recovery.	Tests 3B–3H documented as PASS in uploaded Task 3 report.
4	Communication fault isolation; relay mismatch; battery / ADC faults; recovery ladder; SHUTDOWN.	Older test fixtures plus final integrated relay-mismatch recovery log.
5	Live telemetry; event generation; queue; Wi-Fi reconnection / replay; RSSI; transition data.	Live telemetry and transitions observed. Outage / FIFO replay was tested with temporary verification controls; final build contains no forced-outage commands. Significant-change trigger test remained inconclusive.

Task	Verification areas	Evidence status
6	Risk score; fault count; recent-fault window; maintenance advice; historical charts; Blynk Timeline.	Dashboard and transition logging observed; risk score is a heuristic and chart history requires incoming samples.

11.2 Representative final integrated test

The final integrated relay-mismatch test is a key cross-task demonstration. The system detected a mismatch between relay command and feedback, entered FAILSAFE, retained RELAY_MISMATCH as the active fault and executed the recovery ladder. All four recovery stages passed. The system returned to NORMAL after approximately 4.025 seconds, and both entry and recovery transitions were logged to Blynk. The dashboard subsequently showed one distinct fault and a recent-fault count of one, rather than double-counting the state escalation.

11.3 Build and deployment evidence

The final source was subjected to a static C++ syntax check using lightweight API stubs during consolidation. This check is not equivalent to a full PlatformIO build against the target ESP32 toolchain. The final integrated firmware was subsequently exercised in the Wokwi workflow as reflected by the submitted logs and dashboard screenshots. Before repository submission, the project owner should perform and capture a clean PlatformIO build using the exact final source and dependency configuration.

12. Challenges, Assumptions and Limitations

12.1 Challenges encountered and resolutions

Challenge	Resolution / final treatment
Wokwi made controlled network interruption difficult.	A temporary verification control was used for an offline-queue test; the control was removed from production firmware. A separate SW2 fixture was also used in early Task 4 communication tests and later removed.
Realtime Blynk library connection / Serial interaction caused integration difficulty.	The final firmware uses HTTPS batch updates and a separate network task instead of the persistent realtime Blynk library.
Blynk Timeline endpoint returned HTTP 400.	The BMS State Transition event with code bms_transition was created in the template and Device page visibility enabled; subsequent transition log requests succeeded.
History charts initially showed No data.	History aggregation and a 30-second heartbeat were used to accumulate chart samples; plots only appear after samples arrive.
Fault count could be incremented on state escalation.	Counting was changed to increment only when a new non-NONE fault source / ID appears; the same fault progressing to SHUTDOWN is not counted twice.
The physical input count is limited by direct ESP32 ADC availability.	The software cell loop is parameterised; larger physical configurations require a suitable external ADC or analogue multiplexer and safe input front end.

12.2 Engineering assumptions

- The four potentiometers represent four cell-voltage signals in the simulated system.
- The ADC range is treated as 0–3.3 V and the measured percentage is linearly normalised for demonstration.
- The battery chemistry and chemistry-specific SoC curve are outside the current prototype scope.
- The relay feedback switch represents a binary auxiliary contact; it is not a physical contactor feedback assembly.
- Wi-Fi and Blynk availability are external communication dependencies; the local protection path remains separated from network delivery.
- The offline telemetry queue and rolling fault timestamps are held in RAM and are cleared on restart.
- The 4-to-16-cell memory estimate covers the BatteryCell array only and assumes an 8-byte structure size.

12.3 Limitations and safety considerations

- The Wokwi model is not connected to a real battery pack. Direct high-voltage pack testing has not been performed.
- A production implementation needs appropriately rated voltage dividers / isolated measurement devices, input protection, calibration, filtering and a verified cell-monitoring front end.
- The prototype does not implement active cell balancing; the dashboard can recommend inspection or balancing but no balancing circuit is controlled.
- The adaptive imbalance threshold is SoC-based in the supplied firmware; discharge-rate adaptation is not implemented because no current sensor or load model is present.
- The risk score is a demonstrator heuristic and is not a validated SoH or safety prediction.
- The 24-event FIFO and 16-entry recent-fault window are volatile. Non-volatile storage would be needed to preserve data across reset or power loss.
- The prototype HTTPS client calls setInsecure(); production deployment must validate the server certificate and protect credentials.
- The 16-cell capability is a software architecture analysis, not a completed physical 16-cell hardware build.
- The project is a learning / evaluation prototype and must not be used as the sole protection system for an actual battery pack.

13. Conclusion and Future Improvements

The project integrates six BMS functions into one ESP32-based simulation: modular cell analysis, non-blocking relay protection, flicker-free LCD display, supervisory fault-state management, event-driven telemetry and Blynk analytics. The modular battery engine maintains cell-level information and calculates pack imbalance and adaptive thresholds. The relay protection state machine filters transient faults and performs controlled recovery. The supervisory FSM preserves fault identity and requires sequential verification before clearing FAILSAFE. The display prioritises safety messages, while the network task buffers events and sends state and battery data to Blynk.

The available verification evidence demonstrates the expected behavior of the core monitoring, protection, display and supervisory functions. The final integrated relay-mismatch recovery test confirms the ADC-to-communication recovery ladder and Blynk transition logging. The analytics dashboard demonstrates live status, risk score, fault count, recent-fault information, maintenance advice and historical traces. Some tests, particularly the significant-change trigger and a controlled outage on the final production build, remain qualified in the evidence record and are not overstated as fully verified.

13.1 Future improvements

- Integrate a battery-monitoring AFE or external ADC channels for sixteen or more physical cells, including properly rated and isolated measurement circuitry.
- Use chemistry-aware SoC estimation, current sensing, temperature sensing and calibrated per-cell voltage measurement.
- Implement validated cell balancing, contactor drivers, pre-charge and independent hardware protection.
- Persist event queues and fault history in non-volatile storage or a remote durable backend.
- Validate TLS certificates and move cloud credentials into secure configuration storage.
- Validate the risk score with measured battery datasets and clearly defined operating envelopes.
- Perform hardware-in-the-loop and real battery pack tests with independent safety supervision.

14. References

[1] Internship task specification and common submission instructions supplied for the Embedded Systems / IoT project.

[2] Task 1 verification document: Modular Battery Management Engine (uploaded PDF, 7 pages).

[3] Task 2 verification document: Non-Blocking Protection Relay and Safety System (uploaded PDF, 12 pages).

[4] Task 3 verification document: Flicker-Free LCD Display Engine (uploaded PDF, 5 pages).

[5] Task 4 verification document: Fault State Machine and Structured Recovery (uploaded PDF, 7 pages).

[6] Smart_BMS_Final_Integration_Guide.md, final Task 5–6 integration notes supplied with the consolidated firmware.

[7] Smart_BMS_Final_main.cpp, consolidated ESP32 firmware source (Tasks 1–6).

[8] Blynk Console template configuration and dashboard screenshots supplied during the project verification.

Appendix A - GPIO and Interface Map

Signal / peripheral	GPIO / interface	Notes
Cell 1–4 analogue inputs	34, 35, 32, 33	Four potentiometer inputs in the Wokwi circuit.
LCD I2C	SDA 21, SCL 22	I2C address 0x27; 16 × 2 LCD.
Relay output	19	Software-controlled relay state output.
Relay feedback	27	Simulated auxiliary-contact switch; pull-down input.
Red / green / yellow LEDs	2 / 4 / 5	Status indicator outputs.
Buzzer	18	Fault indication output.
Wi-Fi	ESP32 internal radio	Wokwi-GUEST in the simulation.

Appendix B - Blynk Datastream Map

Pin	Datastream	Type	Meaning
V0	Average voltage	Number	Pack average voltage (V)
V1–V4	Cell 1–4 voltage	Number	Individual cell voltage (V)
V5	Weakest cell	String	Cell label for lowest-voltage cell
V6	Strongest cell	String	Cell label for highest-voltage cell
V7	Voltage imbalance	Number	Maximum minus minimum voltage (V)
V8	Battery status	String	NORMAL / IMBALANCE / WEAK CELL / HIGH VOLT
V9	Battery SoC	Number	Prototype estimated SoC (%)
V10 / V11	Minimum / maximum voltage	Number	Cell minimum and maximum (V)
V12	Adaptive limit	Number	Current imbalance threshold (V)
V13	Imbalance trend	String	START / RISING / FALLING / STABLE
V14	System state code	Integer	0 NORMAL, 1 DEGRADED, 2 FAILSAFE, 3 SHUTDOWN
V15	Relay state code	Integer	Relay FSM state code
V16	Active fault ID code	Integer	Numeric fault identifier
V17	Wi-Fi RSSI	Integer	Received signal strength (dBm)
V18	Offline queue depth	Integer	Remaining event count after successful send
V19	Battery risk score	Integer	Prototype heuristic score, 0–100
V20	Maintenance advice	String	Rule-based recommendation
V21	Fault count	Integer	Total distinct fault occurrences
V22	Dropped events	Integer	Events dropped due to queue overflow
V23	Device uptime	Integer	Uptime at event capture (seconds)
V24	Sent events count	Integer	Successful telemetry transmissions
V25	System status	String	Readable supervisory state
V26	Relay state	String	Short relay state text
V27	Latest event	String	Event reason and sequence
V28	Telemetry mode	String	LIVE / REPLAYED

Pin	Datastream	Type	Meaning
V29	Event age	Integer	Age of event when delivered (seconds)
V30	Event sequence	Integer	Monotonic event sequence number
V31	Fault source	String	BATTERY / ADC / RELAY / COMMUNICATION / NONE
V32	Fault ID	String	Readable fault identifier
V33	State transition detail	String	Previous/new system, relay and fault values plus sequence
V34	Wi-Fi manager state	String	CONNECTING / CONNECTED / RETRY_WAIT
V35	RSSI quality	String	NO_LINK / VERY_WEAK / WEAK / FAIR / GOOD / EXCELLENT
V36	Recent faults (10 min)	Integer	Distinct fault occurrences in rolling ten-minute window

Appendix C - Repository and Submission Checklist

- ☐ Publish the final source code and the integrated Wokwi project files in a public GitHub repository.
- ☐ Add a README.md with project summary, setup, Wokwi link, repository structure and Blynk configuration notes.
- ☐ Include BUG_LOG.md, ASSUMPTIONS.md and SELF_ATTESTATION_CHECKLIST.md.
- ☐ Include the system architecture and system workflow diagrams as separate PNG files.
- ☐ Include dashboard screenshots and selected Serial Monitor test evidence.
- ☐ Add the public Wokwi project URL to the README and report; verify it opens without login.
- ☐ Prepare a 2–5-minute demo video in MP4, WebM or MOV format, maximum 100 MB.
- ☐ If the submission form requires a Google Drive dataset link, mark it not applicable or clarify with the coordinator if no external dataset was used.
- ☐ Before final one-time submission, compile the exact final source, review all links and watch the demo video once.